

Legal Requirements Compliance using NLP and Knowledge Graphs

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Abstract—Different new directives on information and software technologies have been recently published by the European Union, such as the Artificial Intelligence Act (AI Act), the Cyber-Resilience Act, the Network and Information Security Directive 2 (NIS2), and the Digital Service Act. Since the enactment of the General Data Protection Regulation (GDPR), the legal compliance have been performed with expensive certifications and reviews made by consultants of various documents (e.g. the Data Protection Assessment), but now the use of new technologies might accelerate the compliance process by using tools to transform complex legal texts into machine-readable knowledge representations. In our work, we construct the knowledge graphs from the regulatory texts and other relevant documents (such as the Software Requirements Specification) and we aim at assessing the compliance by identifying matches between the two graph representations, with the development of an auditor-oriented compliance tool. We plan to leverage on large-language models (LLMs) to assist in aligning requirement specifications across multiple regulatory frameworks. By highlighting the pitfalls of diverse tools during the experimental analysis, we aim to emphasize the necessity of refining legal text processing workflows to enable transparent, efficient, and legally sound compliance report.

Index Terms—Requirements Validation, Legal Compliance, Information extraction, Knowledge Graphs, EU Directives, AI Act

I. INTRODUCTION

The European Union digital strategies have led the publication of many new directives on information and software technologies: the Artificial Intelligence Act (AI Act), the Cyber-Resilience Act, the Network and Information Security Directive 2 (NIS2), and the Digital Service Act. Since the enactment of the well-known General Data Protection Regulation (GDPR), the legal compliance have been performed with expensive certifications and reviews made by consultants of various documents (e.g. the Data Protection Assessment). Manual assessments performed by auditors (legal and technical experts) are expensive and time-consuming [1] and they need tools able to deal with the complex structure of legal

texts. Moreover, organizations face legal consequences and significant penalties in the event of non-compliance: they are frequently expressed as a percentage of annual global revenues (up to 10% in the case of the DMA, 6% under the DSA and AI Act, and 2.5% under the Cyber Resilience Act). AI raises concerns not only for data protection but also transparency [2], fairness [3], information privacy, and freedom of expression [4]. Consequently, the AI Act is the world's first statutory law proposal for regulating AI systems, and it aims to ensure that AI systems in the EU are safe and respectful of fundamental rights and values. Its application extends beyond EU-based organizations and concerns any AI provider, importer, distributor, or authorized representative within the EU.

The AI Act [5] outlines four distinct risk categories and sets specific requirements accordingly. For systems deemed to pose an unacceptable risk, which are outright prohibited, the Act provides explicit examples and exceptions, including the utilization of real-time remote biometric identification in public spaces (such as facial recognition), social scoring systems (classifying individuals based on behaviour, socio-economic status, or personal characteristics), and the use of subliminal manipulation techniques targeting specific vulnerable groups. High-risk systems are permitted, but due to their ability to negatively affect safety or fundamental rights, they must comply with multiple requirements and undergo a compliance assessment throughout their life cycle, both before and after deployment. Examples of high-risk systems include those related to critical infrastructure management, systems in hiring processes or employee ratings, credit scoring systems, and systems with critical impact on law enforcement and interpretation of law. Companies developing or deploying high-risk AI systems must comply with various requirements, including having an appropriate risk management system, logging capabilities, and human oversight.

Instead of treating legal compliance as an after-thought, we

aim at helping software engineering in designing with compliance in mind, using new technologies to transform complex legal texts into machine-readable knowledge representations, that can be used to inform the requirements engineering process but it can also help auditors in verifying legal compliance.

In this work, we use Natural Language Processing tools to extract knowledge graphs from the legal text, using the The AI Act as a case study; we propose a compliance system architecture that constructs relevant legal knowledge graphs from EU legislations and system requirement specifications to compare and search parts of the two graphs for the compliance assessment. The methodology enables reasoning about legal obligations, permissions, and prohibitions in a knowledge representation that can be easily visualized, analysed, and understood by legal experts. Though the rise of large language models (LLMs) launches new opportunities in legal applications [6], such as document drafting, but the use of LLMs in legal application is critical since it needs manual verification because of the hallucinated content generation and dependence on updated information, as new related laws are frequently published.

Section II explains the our research plan, Section III presents preliminary experiments, Section IV describes related work. Section V concludes the paper with some future directions.

II. METHODOLOGY

The work of Amaral et al. [3] addressed an automated support for verifying the compliance of Data Processing Agreements (DPAs) against GDPR by defining DPA-related compliance requirements as “shall” requirements. Legal experts are also accustomed with this format as “shall” is frequent in drafting requirements. The example of [3] derives an excerpt from Data Processing of agreements verified again GDPR requirements where DPA requirements are often declarative which adds more content required for GDPR compliance. There are some relevant and irrelevant parts of both the regulatory texts.

- The relevant parts stand out the requirement similarities between both the directives. In the other side, even if one agreement satisfies both the regulations, there might be some non-related sentences in the agreement that minimizes the overall semantic similarity, and thus obstructs compliance verification.
- The irrelevant part defines that there are no similarities between two legal documents. Therefore, they are not compliant with each other. The non-relevance describes if an agreement is about to get a written approval with sub-processors, while another agreement concerns acquiring the same but from the controller before involving sub-processors. Though there is word similarities between the agreements.

Similarly, verbose legal texts like DPA often require some well-designed mechanisms to make them clearer and more comprehensive. Therefore, we are focusing on the construction of a compliance strategy that will be capable of tackling those

large texts into a structural format by leveraging different tools and frameworks.

A. Research Plan

Our objective is to support experts in legal audit and certification to better audit and ensure the alignment with legal requirements.

1) *Legal knowledge graph*: A legal knowledge graph is a organized representation of legal knowledge comprising interconnected entities, relationships, and attributes. This innovative tool serves as a dynamic repository designed to capture the intricate web of relationships between legal entities, concepts, and principles. The knowledge graph technology is a knowledge organization form with triples as the basic unit, is able to efficiently transform multi-source heterogeneous information into a knowledge representation form close to human cognition. Legal knowledge [7] involves multidimensional heterogeneous knowledge such as legal provisions, judicial interpretations, judicial cases, and defences, which requires extremely high relevance and accuracy of knowledge.

2) *Proposed Architecture*: Amaral et al. [3] developed a tool for GDPR compliance, extracting 45 compliance requirements for DPA in the form of subject-predicate-object structure, categorizing the subjects and objects as participant, predicates or actions as event for semantic frame representation of the legal requirement sentences. Their seminal work, at that time, used technologies that have now been surpassed by LLMs (for natural language processing) and knowledge graphs (for information representation) and their scope was bounded to the GDPR legislation. We aim to develop a compliance tool for others ICT-related EU directives and legislations with more modern technologies.

Figure 1 defines the proposed architecture. The data might be structured or unstructured. Structured data represents organized information in a defined schema, making it easily searchable. In the other side, unstructured data doesn't have a predefined, making it harder to analyse.

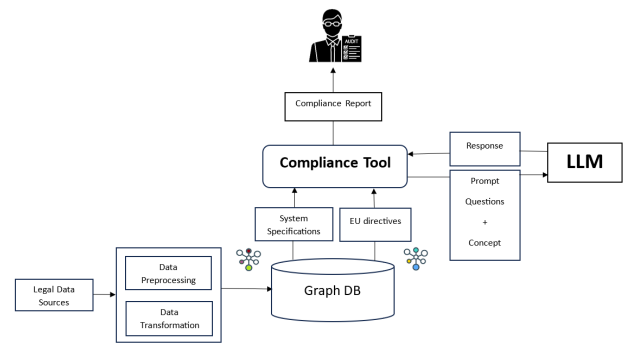


Fig. 1: Overall architecture

Generally, the regulatory texts follow structured format. The legal data sources can vary and depend on the corresponding organizations. The legislative texts from those legal sources can be pre-processed and transformed into graph based

representations. In this case, NLP integrated tools help to extract contractual information from the data source. Large scale Graph Databases assist to create and store multiple graph data structures derived from the legal texts. The graphs are identified for the classification of multiple requirement frameworks to check the compliance similarities. In our case, the compliance tool will compare two graphs, one representing regulatory texts and the other representing a system requirements specification, to determine whether a specific requirement in the SRS is compliant with the corresponding regulatory requirement. At the end, The Tool will be able to generate the compliance report that will showcase the overview whether the compliance tool can pass both the agreements with a detailed representation of possible violations and warnings, similarity scores between the acts to respond the legal auditor inquiries. The compliance tool will later integrate the GraphRAG methods which are most exciting applications of Gen-AI approaches and LLMs for better generation of compliance reports by question-answering through the prompts and related concepts making the representation more effective and interpretable.

III. PRELIMINARY EVALUATION

This section presents how we can use the NLP integrated graph based approaches for our experimental analysis. We use SPRING¹ (Symmetric PaRsIng aNd Generation) [8], a state-of-the-art sequence-to-sequence model for parsing a sentence into an Abstract Meaning Representation (AMR) i.e., a graph-based sentence-level & knowledge-based meaning representation meaning representation heavily relying on frame semantics for linking predicate frames and entity knowledge bases such as DBpedia for linking named entity concepts, and, vice versa, generating a sentence from a given AMR graph, without many cumbersome external components. SPRING exploits the transfer learning capabilities of BART [9]. Such configuration aids data pre-processing by utilizing features such as lemma, POS, or Named Entity Recognition (NER) tags [10]. AMR aims at large scale human annotation in order to build a giant semantics bank. It is also an useful technique that captures many aspects of meaning in a single simple data structure. We employ Neo4j, one of the world's leading graph database, for the knowledge graph creation, as it offers an efficient, highly scalable, reliable and secure way to store, manage and query large amounts of interconnected data. The powerful native graph storage and processing engine, combined with the Cypher query language, makes it easy to quickly build, deploy and maintain a graph database, no matter the size of the project. For our workflow, we use SPRING to convert text into Abstract Meaning Representation (AMR) graphs. We perform Text-to-AMR using SPRING. Text-to-AMR architecture reduces complexity by disposing of the need for content-modifying pipelines and additional syntactic and semantic features. In SPRING, AMR graphs are symmetric in nature: for Text-to-AMR parsing the encoder-decoder is

¹<https://github.com/SapienzaNLP/spring>

trained to predict a graph given an input sentence. Afterwards, we transform the AMR to knowledge graph for creation of Neo4j-built graph representations. Thus, we raise some possible constraints of AMR by focusing on the knowledge graphs.

A. Graph Linearizations

The linearization techniques are fully graph-isomorphic, i.e., it is possible to encode the graph into a sequence of symbols and then decode it back into a graph without losing adjacency information. The AMR graph [8] is a directed graph in a simple, tree like form. The AMR graphs capture predicate argument structure where the root nodes often present predicates. The sentence “the preceding general principles inform the exercise of the discretion” is represented as:

(z0 / inform-01 :ARG0 (z1 / principle :ARG1-of (z2 / general-02) :ARG1-of (z3 / precede-01)) :ARG2 (z4 / exercise-01 :ARG1 (z5 / discretion)))

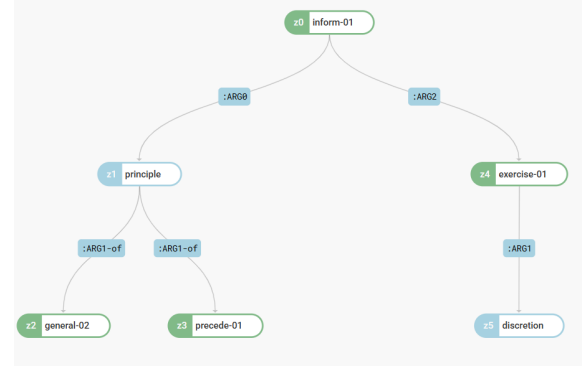


Fig. 2: The AMR graph for the sentence “the preceding general principles inform the exercise of the discretion.” with the linearization

B. Knowledge Graph representation using Neo4j

We conduct a graph transformation approach, converting SPRING-generated AMR properties into the knowledge graph. We use ChatGPT-4o to extract the AMR properties from the AMR graph in terms of nodes and edges to translate them into csv format, supported by Neo4j, through a prompt as:

From the following AMR linearization, extract the nodes and edges into a Python dictionary format. The output should have two keys: 'nodes' and 'edges'. For each node, use its concept name (e.g., 'inform') as the key. Under 'nodes', include its PropBank name (e.g., 'inform-01') as 'name' and assign a coarse semantic type (e.g., 'verb', 'noun', 'adj') based on the concept. Under 'edges', represent relations as 3-tuples (source name, target name, relation), using lowercase relation names. Input: (z0 / inform-01 :ARG0 (z1 / principle :ARG1-of (z2 / general-02) :ARG1-of (z3 / precede-01)) :ARG2 (z4 / exercise-01 :ARG1 (z5 / discretion)))

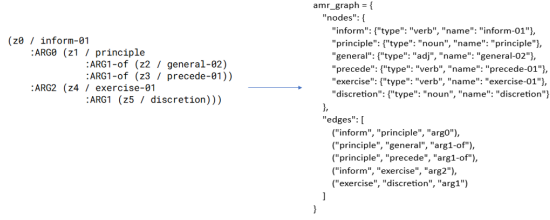


Fig. 3: Graph properties from AMR linearization

1) *Implementation:* We implement the extracted properties to write the CSV files to load in neo4j by using csv library in python to transform them into knowledge graph.

TABLE I: CSV file of extracted nodes

id	type	name
inform	verb	inform-01
principle	noun	principle
general	adj	general-02
precede	verb	precede-01
exercise	verb	exercise-01
discretion	noun	discretion

TABLE II: CSV file of extracted relationships

from	to	type
inform	principle	arg0
principle	general	arg1-of
principle	precede	arg1-of
inform	exercise	arg2
exercise	discretion	arg1

We use the following cypher query to create nodes and relationships from the CSV files: 'id' denotes the PropBank [11] names and 'name' for node names, 'id' denotes source and destination nodes in case of relationships, 'type' denotes the parts of speech of the nodes. The *apoc.create.relationship* calls the APOC procedure² that allows creating dynamic relationships between two nodes.

```

For creation of nodes:
CREATE (n:NODES { id:$id, type:$type, name:$name })

For creation of relationships:
MATCH (a:NODES {id: $from_}), (b:NODES {id: $to})
CALL apoc.create.relationship(a, $relationship,
{ }, b)
YIELD rel
RETURN rel

```

Fig. 4: Cypher query to create nodes and relationships

²APOC in Neo4j <https://neo4j.com/labs/apoc/>

Figure 5 illustrates the Neo4j generated knowledge graph from the AMR Graph linearization. Both Figure 2 & Figure 5 illustrate the similar nodes of AMR and KG. The code is available in github³. The Neo4j Desktop application provides a user-friendly knowledge graph visualization as query outputs. Hovering or single clicking on a node shows its relative property key values on the right-hand side detail box.

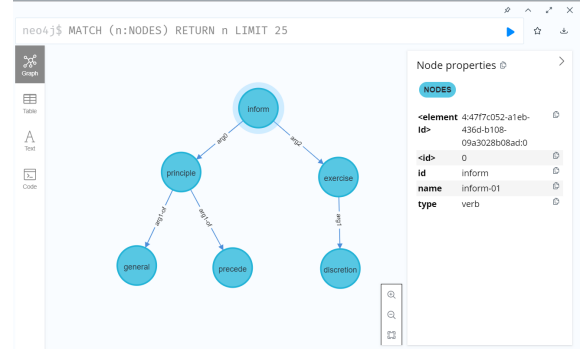


Fig. 5: Knowledge graph from the AMR properties in Neo4j Desktop with its graph generation Cypher query

C. Discussion

From the earlier sections, AMR graphs offer a vast representation of regulatory texts with several fragments by introducing a predicate-argument structure within a graphical representation. While these kinds of structure capture deep semantic roles effectively, they produce relatively higher node count due to the inclusion of auxiliary concepts, and nested events. These granular presentations result redundancy and complicated scalability for the compliance system foundation in spite of linguistically insightful. The transformation from AMR-to-KG illustrates complexity due to structural mismatches and the lack of a unified schema between AMR and KG due to the AMR's semantic role-labeled edges (e.g., :ARG0, :ARG1, :mod). AMR-derived graphs tend to maintain their verbose structure, limiting their effectiveness for high-level compliance reasoning. For the partial evaluation, We utilized ChatGPT-4o for the legislative knowledge graph construction providing legal texts as related concepts by using a prompt as:

"Your task is to give me the triples from the following text by integrating SPO format and create the knowledge graph: 1. The purpose of this Regulation is to improve the functioning of the internal market and promote the uptake of human-centric and trustworthy artificial intelligence (AI), while ensuring a high level of protection of health, safety, fundamental rights enshrined in the Charter, including democracy, the rule of law and environmental protection, against the harmful effects of AI systems in the Union and supporting innovation.

2. This Regulation lays down:

(a) harmonised rules for the placing on the market, the putting into service, and the use of AI systems in the Union;

³<https://github.com/PurbashaChowdhur/AMR2KG>

- (b) prohibitions of certain AI practices;
- (c) specific requirements for high-risk AI systems and obligations for operators of such systems;
- (d) harmonised transparency rules for certain AI systems;
- (e) harmonised rules for the placing on the market of general-purpose AI models;
- (f) rules on market monitoring, market surveillance, governance and enforcement;
- (g) measures to support innovation, with a particular focus on SMEs, including startups.”

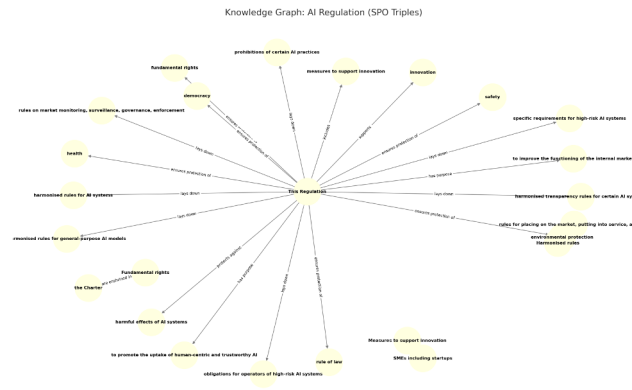


Fig. 6: Generated KG from AI ACT Article 1 by ChatGPT-4o

Many developers try to build retrieval-augmented generation (RAG) experiences to interact with information from unstructured data using only vector search and struggle to get the expected results. Looking only at text fragments without context only gets one so far. As usual in data engineering, there are more advanced patterns for preprocessing the data and extracting knowledge, one of which is GraphRAG [12]. So, when one gets around using the data, they have surfaced the underlying concepts and can make use of them to connect the pieces and provide relevant context to a user’s questions. We used Neo4j’s LLM Knowledge Graph Builder which takes unstructured data, preprocesses documents, transcripts, web articles, and more sources into chunks, compute text embeddings, and connect them (lexical graph). It also enables extraction of entities and their relationships, which is especially relevant if one ingests multiple documents because it can relate the pieces spread out over multiple sources (entity graph). This combined knowledge graph then allows a set of different retrievers to fetch data. The knowledge graph from the same article of AI ACT by Neo4j’s LLM knowledge graph builder is shown in Figure 7: yellow for policy nodes, red for rule nodes, green for objective nodes, blue for activity nodes, and purple for organization nodes. Due to constraints in Neo4j, some node labels are abbreviated (e.g., ‘specific requir...’ for ‘specific requirements for high-risk AI systems and obligations for operators of such systems’). Therefore, unlike AMR, knowledge graphs follow a specific subject-predicate-object (SPO) form using predicates on edges instead of separate nodes, resulting in a significantly reduced

node count and a better hierarchical organization of legal content.



Fig. 7: AI ACT Article 1 SPO Knowledge Graph by Neo4j

Neo4j also supports the incorporation of a chunking strategy for processing large files i.e., the construction of knowledge graphs for individual data segments by maintaining a hierarchical representation where the structure will follow the path to reach to the corresponding articles including related recitals and annexes. Figure 8 depicts knowledge graph where the node risk AI system directs some related articles connected by the outgoing edges.



Fig. 8: Relationship Between High-Risk AI Systems and some Relevant Articles in the EU AI Act

Thus, we aim to create knowledge graphs from EU legislation e.g., AI ACT, NIS2 to verify their requirements in agreement with other software specifications which enable reasoning about legal obligations, permissions, and prohibitions in a knowledge representation that can be easily visualized, analysed by legal experts. By aligning or matching nodes and relationships between the system specification graph and the legal graph, we can identify compliance gaps, potential violations, or confirm alignment. For example, specific obligations extracted from the AI Act can be directly checked against a system’s security or data management requirements.

IV. RELATED WORK

Regulatory compliance is widespread in the RE literature [13]. In RE literature, introducing requirements in regulations relies extensively on model-driven engineering [14]. Several research works have been proceeding with similar methods for representing compliance requirements over the decades. Breau

et al. [15] extract and prioritize rights and obligations from the US healthcare regulation and represent them by applying semantic parametrization that enables expressing rights and obligations in restricted natural language statements. Amaral et al. [16] presented an approach for DPAs compliance against GDPR. Binder et al. [17] presented an approach to automate the privacy threat modelling using static code analysis and extended it for GDPR compliance checks. Abualhaija et al. [18] presented COREQA to assist requirements engineers in acquiring a better notion of compliance requirements by means of automated Question Answering. Cejas et al. [19] introduced a tool named CompAI for checking the completeness of privacy policies to check their compliance to GDPR requirements. Ingolfo et al. [20] presented Nomos2 to model law-compliant solutions in software system design by providing a graphical notation to visualize models and tool support for compliance analysis.

The intersection of Natural Language Processing (NLP) and Legal Knowledge Representation has increasingly become a crucial topic, specifically in automating regulatory compliance. Cabot et al. [21] have proposed a method to verify GDPR compliance in data processing agreements using NLP to compare contractual language with mandatory legal provisions. Similarly, Lippi et al. [22] introduced a new framework that uses machine learning and NLP techniques to evaluate legal compliance by automatically extracting processes for obligations and constraints from the regulatory texts. Ontology-based compliance models have also been considered as a means of converting legal requirements into a machine-readable one [23]. Chennu et al. [24] extracted legal entities, relationships, and constraints using natural language processing (NLP) and ontology engineering which are organized into an ontology and converted into a machine-readable knowledge graph, allowing manufacturers and stakeholders to visualize and get clear information about legal requirements easily.

RAG techniques [25], [26] are widely adopted in recent research works and useful to transform the large scaled data into graphs. They handle long texts by structuring them into a graph.

The use of Large language models simplify the extraction of the context from the legal texts. BERTopic [27] creates dense clusters allowing for easily interpretable topics whilst keeping important words in the topic descriptions. Automated and semi-automated approaches combined with NLP [28], also human-driven approaches across crowd sourcing [29], have been explored in the RE literature to classify privacy-compliant textual content. Legal-Bert, a language model specialized for legal text [30], is a significant development in this field. [31] applied Bert-based approach to assess GDPR compliance in the web service policies, demonstrating how transformer models can enhance accuracy in legal text classification. In [32], the data was collected from legal citations in judicial rulings of U.S. case law where the analysis is of how logical relations are represented in abstract meaning representation (AMR). Bevilacqua et al. [8] introduced a seq2seq approach, SPRING (Symmetric PaRsIng aNd Generation), where model

does not require complex pipelines, nor the need of graph re-categorization. seq2seq-based models include features such as lemma, POS, or Named Entity Recognition (NER) tags, syntactic and semantic structures [33]. Kapanipathi et al. [34] proposed a modular KBQA system, that leverages AMR parsers for task-independent question understanding and an effective graph transformation approach to convert AMR parses into candidate logical queries that are aligned to the Knowledge base.

In recent times, automated knowledge graph construction is achieved by developing LLMs [21]. To reduce the need for human interaction, researchers have tested zero-shot prompting techniques, which means LLMs like GPT-3.5 are guided to identify all the legal key entities and relationships from the text itself. Besides, knowledge graphs also play crucial role in both of software [35] and requirement engineering fields. Karras et al. [36] examined the use of the Open Research Knowledge Graph (ORKG) to build and publish an initial Knowledge Graph of Empirical research in RE (KG-EmpIRE). Colombo et al. [6] presented an investigation of how Legislative Knowledge Graphs and LLMs can support legislative processes by addressing the features of Knowledge graphs for legislative systems, accuracy of large language models (LLMs) for legislative processes, and how non-experts can effectively utilize these tools, mostly focusing on the Italian legislative system.

In contrast, our methodology integrates EU directives and system requirement specifications to capture obligations, permissions, and prohibitions in a semantically rich, legally grounded knowledge graph. This approach enables more accurate compliance modelling and reasoning compared to traditional methods.

V. CONCLUSION & FUTURE WORK

In this work, we outlined the research preview focusing on construction of a compliance approach by using different European directives and legislation and leveraging diverse NLP frameworks. Our method simplifies the compliance verification to assist the auditors for better complying the requirements of the regulatory texts. Although the implementation and evaluation are ongoing, the initial design and insights suggest strong potential for a cutting-edge development. In future works, We will take facilities of legal ontologies to obtain the similarities between the legal entities of two different legislative knowledge graphs by leveraging the functionalities of a pre-trained language model or legal NLP methods. We plan to conduct systematic evaluations with regulatory documents and the original software requirement specifications. We will verify the correctness of graph alignments and assess practical usability by involving legal auditors and domain experts. Additionally, we aim to measure the accuracy and completeness of compliance assessments, thus providing concrete evidence of the approach effectiveness and scalability. We plan to utilize node embedding approaches to measure the similarities and combine node features. We will conduct comparative analysis between the neo4j generated legal knowledge graphs with the

others LLM or NLP-integrated tools built legal knowledge graphs. We plan to leverage GraphRAG or Gen-AI approaches to enable question-answering capability of our mechanism.

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